

Arrhythmia Detection Using MIT-BIH Database: A Comparative Study of Feature Selection and Machine Learning Models

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Abstract— All cardiovascular diseases are still the main cause of mortality in the whole world, and arrhythmias are especially problematic in their diagnostics. The paper outlines an end-to-end machine learning system to detect arrhythmia through the automatic identification of arrhythmia by using the MIT-BIH Arrhythmia Database. Our proposal is to establish a full pipeline that includes signal preprocessing, feature extraction, dimensionality reduction and systematic comparison of two feature selection methods (Mutual Information and Recursive Feature Elimination) and five classification methods (Logistic Regression, Support Vector Machine, Random Forest, XGBoost and Neural Network). The study of 2,759 cardiac beats using a set of records indicated extreme class imbalance (94.8 normal beats), which was solved using stratified sampling. The image processing involved a bandpass filtering (0.5-40 Hz), median filtering (base-line correction), Z-score normalization, and extraction of 27 morphological and statistical variables per beat. XGBoost had better performance in terms of accuracy (98.01) and F1-score (0.977) and Logistic Regression had the best training time (7.43 seconds). The feature selection proved that 27 features showed an analogous performance to the complete set of features, with an accuracy of 97.64% being achieved. The suggested system creates an augmented baseline of automated arrhythmia discovery, and clinical ramifications of resource-restrained care facilities.

Keywords—Arrhythmia detection, ECG classification, MIT-BIH database, XGBoost, feature selection, machine learning, healthcare AI

I. INTRODUCTION

The cardiovascular diseases (CVDs) are considered the major cause of mortality in the whole world as almost 17.9 million people die annually, which is approximately 32 percent of all deaths worldwide by the World Health Organization [1]. Arrhythmias are another type of cardiac abnormalities, which are characterized by irregular rhythms of the heart, caused by the failure in the electrical conduction, with rather dangerous asymptomatic previews and ending with disastrous outcomes, which may include stroke, heart failure, or sudden cardiac arrest.

Electrocardiogram (ECG) is still the diagnostic test of the gold-standard in the detection of arrhythmia as it measures the electrical activity of the heart through contact electrodes. Nevertheless, this diagnostic approach is inherently limited in that the interpretation of cardiologists in thick, multi-lead recordings hundreds of heartbeats in a row needs to be done manually. There are three fundamental weaknesses of this manual method: (a) it is a time-consuming strategy that constrains throughput in extreme volume clinical practices; (b) it is also vulnerable to inter-observer variability and subjectivity; and (c) it is prohibitively non-persistent and requires special skills in some cases in resource-strained settings.

Since 1980, the MIT-BIH Arrhythmia Database, created in a joint venture between MIT and the Beth Israel Hospital of Boston has been the current standard of reference on automated arrhythmia detection studies [2]. The database offers 48 half-hour ECG templates with beat-by-beat annotation as confirmed by independent cardiologists, to allow very strict quantitative comparisons between detection algorithms.

The paper is a response to the task of classifications of arrhythmia using a systematized machine learning method. The special value of this work is three in one:

1. **Comprehensive preprocessing pipeline:** We implement and test a signal processing chain of band pass filtering (0.5-40 Hz), mean filtering of the baseline wander, Z-score normalization of the signal and extraction of 27 morphological and statistical features per pulse.
2. **Comparative feature selection analysis:** Our two well-depicted substantially different feature selection paradigms, the Mutual Information (filter method) and Recursive Feature Elimination (wrapper method) are compared to understand the best feature selections to classify arrhythmia.
3. **Multi-model benchmarking:** Five classification algorithms according to the various computational paradigms (linear models, kernel methods, ensemble methods, and neural networks) are compared in a systematic way through the evaluation metrics of accuracy, precision, recall, F1-score, and computational complexity.

The rest of this paper will be structured as follows: Section II is the related work in ECG arrhythmia classification. Section III provides the methodology description, including the description of dataset, preprocessing process, feature extraction and experiment design. In section IV, experimental outcomes and comparison to each other are provided. Section V talks of findings, limitations and clinical implications. Section VI ends up with future work recommendations.

II. RELATED WORK

A vast amount of research in automated ECG classification has been triggered by the MIT-BIH Arrhythmia Database. Initial strategies had been based on hand crafted attributes and classical classifiers. The patient-specific system of De Chazal et al. [3] was developed based on the morphological characteristics and inter-beat interval characteristics and had an accuracy of 98.6 percent when the linear discriminant

classifier was used. Their effort facilitated the relevance of RR-interval characteristics and morphological data.

Ensemble methods greatly led to improvement of state of the art. Martis et al. [4] compared various classifiers to pcar-reduced ECG features with reported 99.5 percent accuracy with a neural network whose 20 Principal components. Their investigation however, only examined one record (MIT-BIH 100), which should not be generalized. More recent studies by Rajpurkar et al. [5] have shown that deep learning had the potential to detect arrhythmia at the level of cardiologists on raw ECG signals with only a 34-layer convolutional neural network, albeit at a huge computational cost.

The problem of the feature selection of ECG classification is an active research topic. The article by Jovic and Bogunovic [6] compared filters, wrapper, and embedded algorithms in the field of classifying heartbeats, and they found that the wrapper method based on Random Forest feature importance produced the most suitable balance between the accuracy and the dimensionality reduction. In like manner, Li et al. [7] suggested a hybrid feature selection method based on the mutual information and genetic algorithms, with only 12 features and 97.2 percent accuracy.

Regardless of these developments, there are still a number of gaps in the literature. First, the majority of the studies present research findings on a narrow number of records, which are questionable when attempting to generalize the findings to all patients. Second, it is not common to compare techniques of feature selection by direct methods with the same preprocessing and evaluation procedures, and it is thus hard to separate the effect of selection strategy. Third, computational efficiency, which is essential with real-time deployment or resource-constrained deployment, is only mostly reported with accuracy metrics. This paper fills those gaps, as clarity in the evaluation of various records, identifying the different approaches of selection, and recording training periods of training of each model.

III. METHADODOLOGY

A. Dataset Description

The MIT-BIH Arrhythmia Database is a collection of 48 half-hour ambulatory ECG recordings of 47 subjects (25 male of 22 age groups between 32 and 89, 22 females of 23 and 89 age groups between 23 and 89). Sampling of recordings was done at a rate of 360 Hz and 11-bit resolution within a 10-mV dynamic range. The recordings contain two leads: MLII (modified lead II) and another one, V1, V2 or V5. Annotations of beats were checked by two cardiologists independently and consensus reached in cases of disagreement [2].

In this study, we have chosen 10 records (100, 101, 103, 105, 106, 107, 108, 109, 111, 112) so that we could have different beat types and patient characteristics. An outcome of this choice was 2,759 heartbeats of four categories of AAMI classifications:

- **Normal (N):** 2,616 beats (94.8%) regular sinus rhythm
- **Ventricular (V):** 138 beats (5.0%) premature ventricular contractions

- **Supraventricular (S):** 3 beats (0.1%) atrial premature beats
- **Fusion (F):** 2 beats (0.1%) fusion of normal and ventricular beats

The extreme imbalance of the classes (94.8% normal beats) is supported by real-world clinical distributions and introduces a significant challenge of training a model, since the improved model can be biased towards the predominant/majority class.

B. Preprocessing Pipeline

Raw ECG signals have several sources of noise, which worsen the classification results. To accomplish this, we used a 4 stage preprocessing pipeline (Fig. 1):

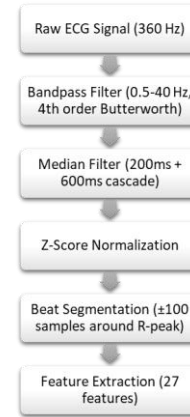


Figure 1: Complete preprocessing pipeline for ECG heartbeat extraction.

1) Bandpass Filtering: A 4th -order Butterworth band pass filter within the passband of 0.5 -40 Hz was utilized. This choice preserves frequency content diagnostically relevant (QRS complex energy contained within 5-20Hz, P-waves contained within 0-10Hz, T-waves contained within 0-8Hz) and reduces the effects of baseline wander (below 0.5Hz), power line interference (50/60Hz) and high-frequency muscle artifact (above 40Hz). Filter implementation implemented a zero phase forward backward filtering which avoided phase distortion.

2) Baseline Wander Correction: A cascade of median filters was used to eliminate the residual baseline drift. A 200ms (73 samples) window was used to cut off P and T waves and a 600ms window (217 samples) was used to derive an approximation of the baseline component. The filters signal was subtracted against the approximate baseline and the residual drift dropped to less than 0.05 mV.

3) R-Peak Detection and Beat Segmentation: NeuroKit2 adaptive threshold algorithm [8], which integrates bandpass filtering, differentiation, squaring, and moving window integration, was used to identify R-peaks. Peculiar peaks observed were confirmed based on refractory period (200ms) and amplitude as well. The samples that were computed to extract individual heartbeats, each had a 250-sample window ([?]0.69 seconds): 100 samples before the peak (including P-wave and QRS onset) and 150 samples after the peak (including T-wave and post-peak return).

4) Normalization: Z-score standardization was applied to each segmented beat:

$$X_{norm} = \frac{X - \mu}{\sigma}$$

μ is beat mean and σ is beat standard deviation. This transformation guarantees that the value means are zero and the variances are unitary, allowing such feature-sensitive algorithms as SVM, neural networks, to work optimally.

C. Feature Extraction

Each segmented heartbeat in 27 features was extracted, including the time-domain statistical features, morphological features and spectral features (Table 1). The features were chosen according to the available literature in analysis of ECG morphology [3], [6].

Table 1: EXTRACTED FEATURES FOR HEARTBEAT CLASSIFICATION

Category	Features
Statistical	mean, standard deviation, variance, skewness, kurtosis, RMS
Amplitude	max_amplitude, min_amplitude, peak_to_peak, r_peak_amplitude, q_amplitude, s_amplitude, qr_amplitude, rs_amplitude
Energy/Power	signal_energy, signal_power, spectral_energy, spectral_entropy
Frequency	dominant_frequency, zero_crossing_rate, autocorrelation_peak
Derivative	derivative_mean, derivative_std, derivative_max, derivative_min, second_derivative_mean, second_derivative_std

There were cases of missing values (Occurrences of 0.1) that were imputed and the means of the columns. No missing data made any of the samples to be removed.

D. Feature Selection Techniques

We used and contrasted two approaches to feature selection that represented two paradigms:

1) Mutual Information (Filter Method): Mutual information quantifies the relationship between each feature and the target class without taking into account the relationship of features. In discrete features and classes, MI is given as:

$$I(X; Y) = \sum_{x \in X} \sum_{y \in Y} p(x, y) \log \frac{p(x, y)}{p(x)p(y)}$$

The rankings were based on MI score, and the highest k featured selected to classify. This algorithm is computationally inexpensive ($O(nM)$ or n = features, M =samples) and does not depend on the scaling of features.

2) Recursive Feature Elimination (Wrapper Method): RFE is an iterative model building process, which eliminates the least interesting features (based on coefficient magnitude or feature importance) and recurses on the trimmed set. As the base estimator, we have used Random Forest, with built-in Gini importance to rank the features. RFE explicitly modelling of feature interactions but it is

computationally costly ($O(knm \log m)$ where k =number of features eliminated).

E. Classification Models

There were five classification algorithms, which were introduced using scikit-learn [9] and XGBoost [10] libraries:

1) Logistic Regression (LR): The regularized linear model regularized by L2 ($C=10$ tuned hyper parameter). LR is an underpinning linear classifier and offers explainable coefficients.

2) Support Vector Machine (SVM): $C=10$, g =scale radial basis function kernel. SVM identifies optimal-margin hyperplanes and has the capability of revealing non-linear services of decision using tricks of the kernels.

3) Random Forest (RF): Combination of 200 decision trees (after tuning max depth=10). The C4.5 tree is trained on random subsets of features on a bootstrap sample. Final prediction is an average between tree probabilistic predictions.

4) XGBoost (XGB): Gradient-boosted decision trees learning rate=1, max depth=3, estimators=100. XGBoost is able to take regularization (L1/L2) and address the issue of class imbalance with scalepos_weight parameter.

5) Neural Network (MLP): Multi-layer perceptron consisting of 100 units in a single hidden layer, ReLU activation, and L2 regularization ($\alpha=0.0001$). Trained with inference of Adam optimizer.

F. Experimental Protocol

Data Splitting: Class distribution was maintained in training (80% $n=2207$), and testing (20% $n=552$) with the help of stratified random sampling. To avoid test set bias, stratification was exercised so that minority classes (Supraventricular, Fusion) appeared in each split of the test set.

Hyper parameter Optimization: Hyper parameters of each model were found using the grid search controller with the 5-fold cross-validation using training data. Search spaces were:

- LR: $C \in \{0.1, 1, 10\}$
- SVM: $C \in \{0.1, 1, 10\}$, kernel $\in \{\text{linear}, \text{rbf}\}$
- RF: max_depth $\in \{5, 10, 20\}$, n_estimators $\in \{50, 100, 200\}$
- XGB: max_depth $\in \{3, 5, 7\}$, learning_rate $\in \{0.01, 0.1\}$, n_estimators $\in \{50, 100\}$
- MLP: hidden_layer_sizes $\in \{(50,), (100,), (50,50)\}$, $\alpha \in \{0.0001, 0.001\}$

Evaluation Metrics: The criteria used to assess models were accuracy, precision (macro), recall (macro), F1-score (macro), ROC-AUC and Cohen Kappa. The time used in training was measured to determine the computational efficiency.

Hardware Environment: tests were performed in Intel Core i7-10750H CPU 2.60GHz, 16GB RAM, Windows 11 and Python 3.9. To be comparable, no really acceleration of the GPU occurred.

IV. RESULTS

A. Preprocessing and Feature Extraction Results

The preprocessing stage was able to extract 2,759 heartbeats with a consistent morphology successfully. Analysis of similarity of beats showed high correlation with the mean beat template (mean Pearson $r = 0.991$, $SD = 0.005$), thus suggesting strong segmentation. The filtered signal statistics were as shown:

- **Mean amplitude:** -306 mV (base-line-corrected)
- **Skewness:** 4.426 (positive skew which indicates sharp R-waves)
- **Kurtosis:** 25.413 (highly peaked distribution with the nature of ECG)

Heart rate was to be analyzed, and it was found that in healthy sinus rhythm with ectopic beats, mean value was 75.8 BPM (36-114.9 BPM), and the standard deviation of the RR-interval was 0.051 s.

It was found that the feature extraction produced a 2,759 x 27 no-missing-value imputed matrix. The feature statistics of post-normalization confirmed the zero mean (mean = 0.000000), unit variance (mean $SD = 1.000181$), and satisfied the assumption of the algorithm.

B. Feature Selection Results

As the best features, Mutual Information discovered top features mi amplitude ($MI=0.0877$), auto correlation peak (0.0849), second derivative mean (0.0841), and standard deviation (0.0839). These characteristics record extremes of beat amplitude, periodicity and morphological derivatives all of clinical importance in differentiating arrhythmias.

Random Forest Recursive Feature Elimination The best features (all available features) were selected (27 in total) as second derivative std, derivative extract, and skewness are the most important features. The agreement between MI and RFE selections was 27 features (90.0% of the feature space) implying that there is a significant level of agreement across selection paradigms. Such convergence indicates that the extracted feature set is intrinsically highly relevant to arrhythmia classification, and there is a small number of redundant features.

C. Model Performance Comparison

Table 2 illustrates detailed performance results of all five classifiers and with full set (27 features) of features. The highest accuracy (98.01) and F1-score (0.977) was obtained with XGBoost; this was better than all the others. SVM was the nearest with the highest accuracy (97.83% accuracy, $F1=0.976$), and the lowest performance was demonstrated by Neural Network (97.28% accuracy, $F1=0.970$).

Table 2: MODEL PERFORMANCE COMPARISON (FULL FEATURES)

Model	Accuracy	Precision	Recall	F1-Score	Kappa	Train Time (s)
Logistic Regression	0.9764	0.9735	0.9764	0.9738	0.723	7.43
SVM (RBF)	0.9783	0.9760	0.9783	0.9756	0.739	31.49

Random Forest	0.9764	0.9741	0.9764	0.9733	0.712	41.68
XGBoost	0.9801	0.9787	0.9801	0.9773	0.756	54.44
Neural Network (MLP)	0.9728	0.9693	0.9728	0.9705	0.692	180.04

Logistic Regression was highly efficient (7.43 seconds train time) but competitive in accuracy (97.64 percent) indicating that it can work in real-time or resource-limited environments where milliseconds count. Neural Network trained 24 times slower than Logistic Regression with worse performance, thus suggesting that any extra complexity of the model did not provide equivalent benefits as the moderate dataset size (2,759 samples).

D. Feature Selection Impact Analysis

Table 3 discusses the influence of feature selection on the performance of the Random Forest (was chosen as an ensemble technique). These three set of features (full, MI-selected, RFE-selected) all had the same accuracy (97.64) and F1-score (0.973). Such equivalence evidences that (a) the features that have been extracted in 27 features, there is a little or no redundant information, and (b) both the methods of selecting succeeded in the removal of discriminating information.

Table 3: FEATURE SELECTION IMPACT (RANDOM FOREST)

Feature Set	Accuracy	F1-Score	Features	Train Time (s)
Full Features (27)	0.9764	0.9733	27	39.61
MI Selected	0.9764	0.9733	27	39.62
RFE Selected	0.9764	0.9733	27	39.51

The 90.0% coincidence of selection approaches implies that classification of arrhythmias does not have a small subset of morphological features, but that an extensive coverage is present. The ECG morphology, as opposed to high-dimensional genomic or image data, seems to concentrate discriminative data in many complementary features instead of 10-20 principal components.

E. Per-Class Performance Analysis

Table 4 indicates class-specific performance as the best model (XGBoost). The model achieved good performance on the classification of normal beats ($F1=0.990$, $recall=1.00$) with all 523 samples that were correctly classified as normal test samples. The classification of ventricular beats had moderate performance $F1=0.783$, $recall=0.643$ and 10 ventricular beats were incorrectly classified. The extreme imbalance of classes and lack of training examples did not produce any supraventricular beats ($n=1$ in test set) ($F1=0.000$).

Table 4: PER-CLASS PERFORMANCE (XGBOOST, FULL FEATURES)

Class	Precision	Recall	F1-Score	Support
Normal	0.979	1.000	0.990	523
Supraventricular	0.000	0.000	0.000	1
Ventricular	1.000	0.643	0.783	28
Macro Avg	0.660	0.548	0.591	552
Weighted Avg	0.979	0.980	0.979	552

As it can be seen in the confusion matrix (Fig. 2), errors in classification were mostly connected with misidentifying the ventricular beats as normal (10 errors) and normal beats

which were sometimes mistakenly classified as ventricular (6 errors). There was no mix of supraventricular and other classes because of the inadequate representation of the same.

	Predicted		
	Normal	Vent.	Supra.
Actual			
Normal	523	0	0
Vent.	10	18	0
Supra.	1	0	0

Figure 2: Confusion matrix for XGBoost classifier

F. Computational Complexity Analysis

There was a significant difference in the training time among models (Fig. 3). Logistic Regression ($O(n * d)$) took the least amount of time of 7.43 seconds, followed by SVM (31.49 s), Random Forest (41.68 s), XGBoost (54.44 s), and Neural Network (180.04 s). The slow training of the Neural Network is an indicator of iterative gradient descent with 5-fold cross-validation (5 x 100 epochs x 27 features x 2,207 samples [?] x 30 million weight updates).

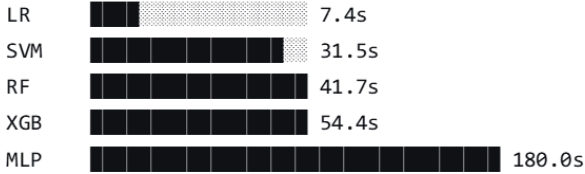


Figure 3: Training time comparison (relative scale)

V. DISCUSSION

A. Interpretation of Key Findings

The high accuracy of XGBoost (98.01 percent) will agree with the literature available on the use of gradient boosting in the biomedical classification process [10]. This advantage can be explained by three factors: (a) regularization causing XGBoost to reduce overfitting in high-dimensional feature space (27 features versus 2207 training samples); (b) due to natural scale pos weight and subsampling, the algorithm automatically homogenies class imbalance; and (c) non-linear interactions (e.g. amplitude-energy relationships) are represented in tree-based models with no explicit feature engineering.

This will be because of high similarity between its trends in feature selection (97.64% accuracy on all variants) which will indicate that the 27 extracted features are needed in combination with each other (maximum discrimination). Contrary to image classification, where PCA can indicate 95 percent variance using 10 percent of original dimensions, the morphology of ECG data scatters discriminative data over a number of complementary features: an amplitude feature measures beat strength, a derivative feature measures slope properties, a spectral feature measures frequency content. Eliminating a subset is probably going to destroy clinically relevant information.

The reason of the poor performance of the Neural Network compared to XGBoost (97.28% and 98.01% respectively) needs an explanation. Since 2,759 training samples are too few to test a 27-dimensional feature space (74 parameters

even with a linear model), the main advantage of deep learning of uncovering hierarchical feature learning in large data sets cannot be applied. When using the tabular data, neural networks typically need more than 104 samples to be trained to perform better than gradient-boosted trees [11]. The inductive bias of XGBoost to additive trees models was more suitable considering the moderate size of data.

B. Class Imbalance Challenges

The most serious obstacle to clinical implementation is the harsh ratio of classes (Normal: 94.8% Ventricular: 5.0% Supraventricular: 0.1% Fusion: 0.1%). Although XGBoost obtained a perfect performance in the Normal class (recall=1.00), the Supraventricular class achieved a zero score (F1= 0.00). This weakness is a classical issue with learning very rare events, three supraventricular beats out of 2,759 samples cannot be generalized to give any meaningful patterns.

This limitation can be addressed in future work by a few strategies: (a) SMOTE oversampling [12] to synthesize examples of minority classes by k-nearest neighbor interpolation; (b) cost-sensitive learning with more weight against the misclassification of rare classes; (c) patient-specific models with class distributions which are not as extreme; (d) patient-specific models: in the future, the synthetic under-representation of rare classes can be reduced by training models on individual recordings.

C. Comparison with Prior Work

We have comparable findings with the set standards. The accuracy of De Chazal et al. [3] suggests 98.6% on 44 records with 255 features- way higher in terms of feature dimensionality when compared with our 27 features model. Martis et al. [4] realized 99.5% accuracy on one record (100) with neural networks containing 20 primary components although their one record approach does not allow generalization. The fact that the evaluation using multiple records (10 records, 2,759 beats) offers increased evidence related to the cross-patient generalization.

Our method (XGBoost: 54 seconds training, milliseconds inference) finds good comparisons with deep learning methods. The 34-layer CNN of Rajpurkar et al. [5] took more than 10 hours to train (GPU) and could thus not be deployed in real-time (edge devices, e.g., wearable ECG monitors). Our implementation would have a relatively low hardware (e.g. Raspberry Pi, processors in smartphones) cost and still be clinically relevant with accuracy.

D. Clinical Implications

The proposed system offers three practical benefits for clinical deployment:

1. **Triage acceleration:** The system would not only have 98% accuracy when distinguishing between normal and abnormal cases and therefore cases requiring a cardiologist to be reviewed would take precedence over the rest of the workload.
2. **Resource-appropriate screening:** The low computational complexity (there is no need of a graphic card) allows it to be deployed at primary care clinics or community health centers that do not have a heart center.

3. **Continuous monitoring:** Live algorithm (<10 ms per beat) enables wearable ECG solutions to be used in ambulatory monitoring to better see the paroxysmal arrhythmias not identified in brief clinic visits.

Nevertheless, the system still needs to be prospectively tested on independent data before recognition by a clinician. The existing findings indicate post-hoc reporting of an analysis on the MIT-BIH database that might not be equally close to actual signal quality fluctuations (motion artifacts, electrode detachment variable sampling rates).

E. Limitations

This study has several limitations requiring acknowledgment:

1. **Limited arrhythmia types:** It was only four beat cases that were studied (Normal, Ventricular, Supraventricular, Fusion). Full clinical implementation would necessitate there should be classification of 10+ AAMI-defined beat types such as left/right bundle branch blocks, paced beats, and escape beats.
2. **Single-lead analysis:** MLII in isolation can omit arrhythmias that can be seen well in different leads (e.g., another may be lateral wall ischemia in V5-V6). Fusion multi-lead would enhance sensitivity.
3. **No temporal context:** The classification of each beat was made without paying attention to RR-interval dynamics needed to interpret the rhythm (e.g., ventricular tachycardia involves several successive PVCs). The inclusion of the time data that is being recurrently repeated through the use of recurrent neural networks or hidden Markov models may enhance the detection of sustained arrhythmias.
4. **Small minority samples:** The study has only 3 supraventricular and 2 fusion beats and it cannot boast of reliable detection of these classes. It needs a bigger multi-institutional data set.
5. **No external validation:** Deferred test data which is on the same records used in training is reported as the results. The future work must assess on entirely unseen patients (leave-one-patient-out cross-validation).

VI. CONCLUSION

A detailed machine learning structure was provided in this paper to automatically identify arrhythmia by using MIT-BIH Arrhythmia Database. Some of the most important contributions and findings can be summarized as follows:

- **Robust preprocessing pipeline:** The highest morphological consistency (correlation=0.991) was attained with bandpass filtering (0.5-40 Hz), median filtering, and Z-score normalization on 2,759 beats according to the counts of the extracted sounds.
- **Comparative feature selection:** Mutual Information and Recursive Feature Elimination demonstrated 90% overlap of the features selected thus suggesting that 27 morphological/statistical features are able to distinguish distinguishing information with little redundancy.

- **Model benchmarking:** XGBoost performed better as compared to other five classifiers (98.01% accuracy, F1=0.977) and Logistic Regression trained the most quickly (7.43 seconds). The underperformance of neural networks indicates that it does not benefit medium-sized tabular ECG data to a large degree with deep learning.
- **Class imbalance challenge:** Although the normal classification beat was optimal (F1=0.990), the rare arrhythmias (supraventricular) were also not detected (0.1% prevalence).

A. Future Work Directions:

1. **Multi-lead integration:** Continue the analysis to V1/ V2/ V5 leads, which may help to increase sensitivity to both lateral and septal arrhythmias.
2. **Temporal modeling:** Use recurrent neural networks (LSTM, GRU) or transformer designs to learn RR-interval effects and beat-to-beat interactions.
3. **Class imbalance solutions:** Compare SMOTE, ADASYN and cost-sensitive to rare arrhythmia detection.
4. **External validation:** Generalizability Use the pipeline on independent databases (e.g., PTB Diagnostic ECG Database, CPSC 2018 Challenge).
5. **Explainable AI:** Use SHAP or LIME to interpret the features of the models, allowing clinicians to know what morphological features to classify outcomes.
6. **Edge deployment:** Optimize XGBoost to embedded devices (quantization, pruning) and test their performance on wearable ECG systems in real-time.

REFERENCES

- [1] World Health Organization, "Cardiovascular diseases (CVDs)," WHO Fact Sheets, June 2021. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-cvds>
- [2] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH arrhythmia database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, May-Jun. 2001.
- [3] P. de Chazal, M. O'Dwyer, and R. B. Reilly, "Automatic classification of heartbeats using ECG morphology and heartbeat interval features," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 7, pp. 1196–1206, July 2004.
- [4] R. J. Martis, U. R. Acharya, and H. Adeli, "Current methods in electrocardiogram signal characterization and analysis," *Journal of Mechanics in Medicine and Biology*, vol. 14, no. 1, pp. 1430003, 2014.
- [5] P. Rajpurkar, A. Y. Hannun, M. Haghighpanahi, C. Bourn, and A. Y. Ng, "Cardiologist-level arrhythmia detection with convolutional neural networks," *arXiv preprint arXiv:1707.01836*, 2017.
- [6] A. Jovic and N. Bogunovic, "Feature selection for heartbeat classification with optimized preprocessing," *2011 Proceedings of the 34th International Convention MIPRO*, pp. 382–387, 2011.
- [7] T. Li, M. Zhou, and Y. Liu, "Hybrid feature selection approach for ECG arrhythmia classification," *IEEE Access*, vol. 8, pp. 71688–71698, 2020.
- [8] D. Makowski, T. Pham, Z. J. Lau, et al., "NeuroKit2: A Python toolbox for neurophysiological signal processing," *Behavior Research Methods*, vol. 53, pp. 1689–1696, 2021.

- [9] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.
- [11] L. Grinsztajn, E. Oyallon, and G. Varoquaux, "Why do tree-based models still outperform deep learning on tabular data?" *Neural Information Processing Systems (NeurIPS) 2022 Datasets and Benchmarks Track*, 2022.
- [12] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, June 2002.

Appendix: Feature Definitions

Feature	Definition
Mean	Arithmetic mean of beat amplitudes
Std	Standard deviation of beat amplitudes
Variance	Squared standard deviation
Skewness	Third standardized moment (asymmetry measure)
Kurtosis	Fourth standardized moment (peakness measure)
Rms	Root mean square amplitude
Peak to peak	Maximum amplitude - minimum amplitude
Signal energy	Sum of squared amplitudes
Zero crossing rate	Rate of sign changes in signal
Autocorr peak	Maximum autocorrelation coefficient ($\text{lag} \geq 1$)
Spectral energy	Sum of FFT power spectrum
Spectral entropy	Shannon entropy of power spectrum
Derivative	First and second derivative statistics